DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; LAMP, SOLID STATE, FOR USE AS REPLACEMENT FOR COMMERCIAL F20T12 FLUORESCENT LAMP IN VITAL AND EMERGENCY LIGHT FIXTURES

This specification is approved for interim use by the Naval Sea Systems Command, Department of the Navy. Other activities in the Department of Defense may use this interim document.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

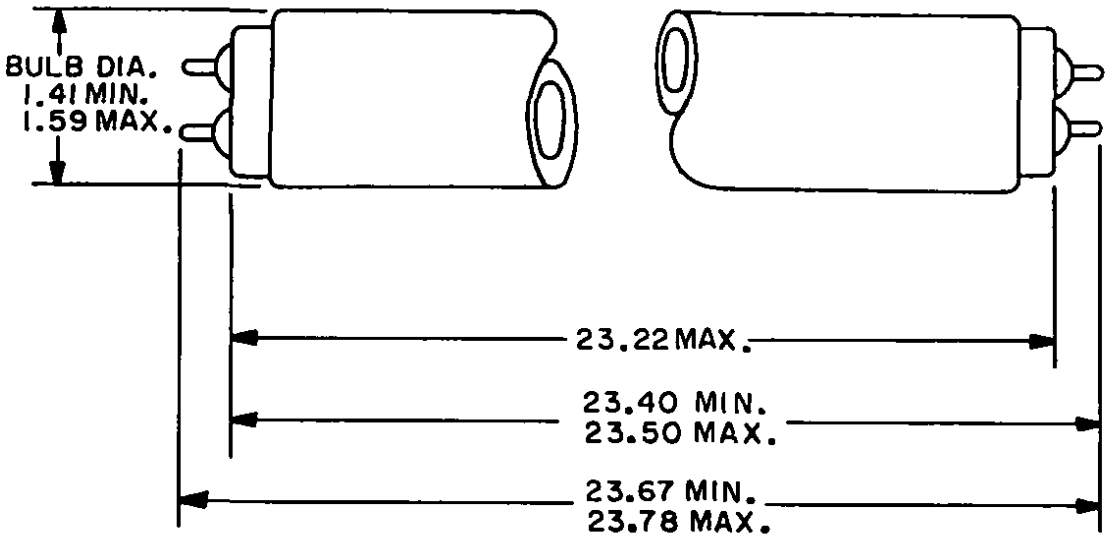


FIGURE 1. Lamp dimensions (inches).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.

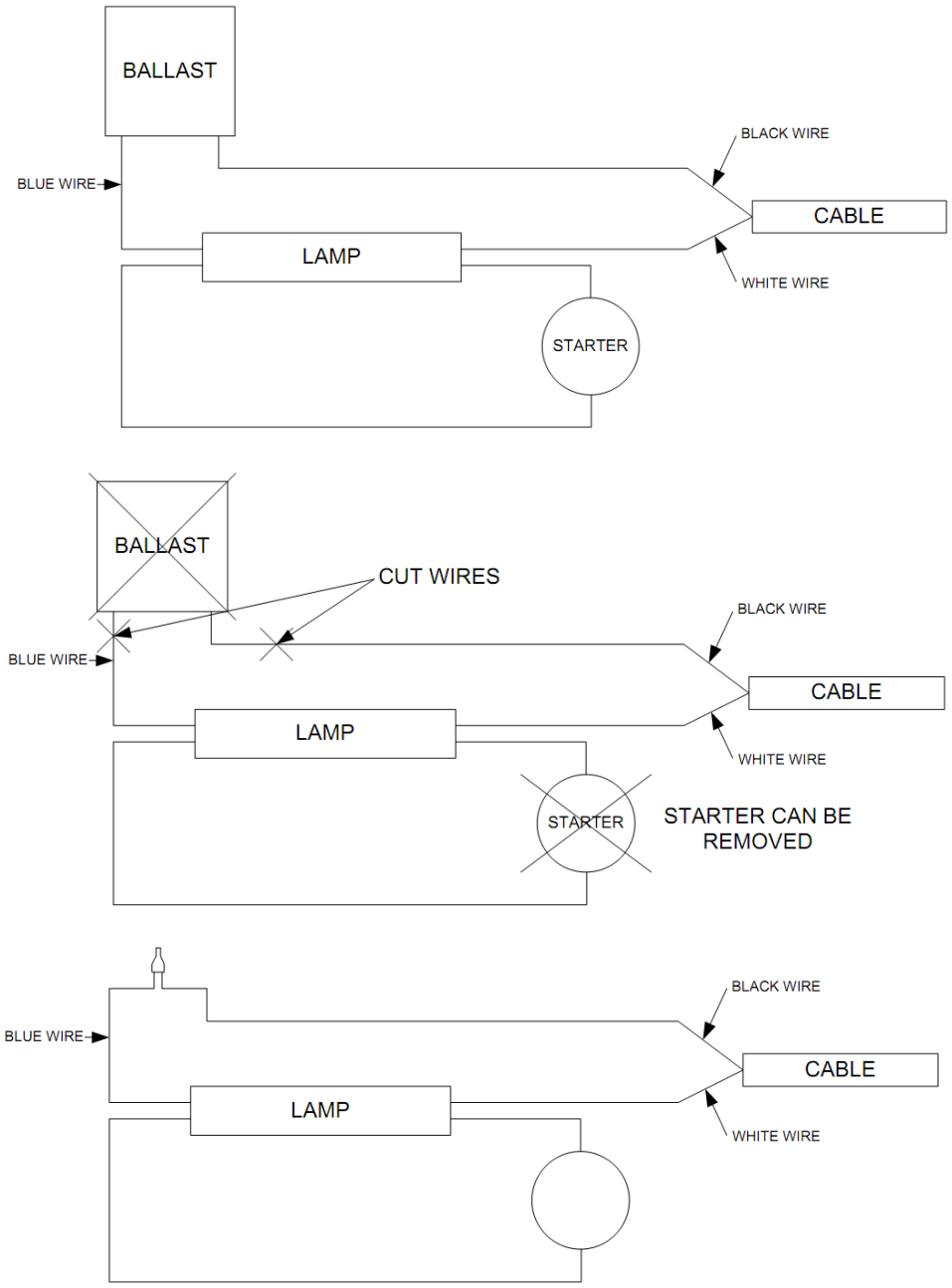


FIGURE 2. Ballast and starter removal for fixtures utilizing M16377/44, 2-wire ballast for 115 volts alternating current (VAC) operation of solid state lamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.

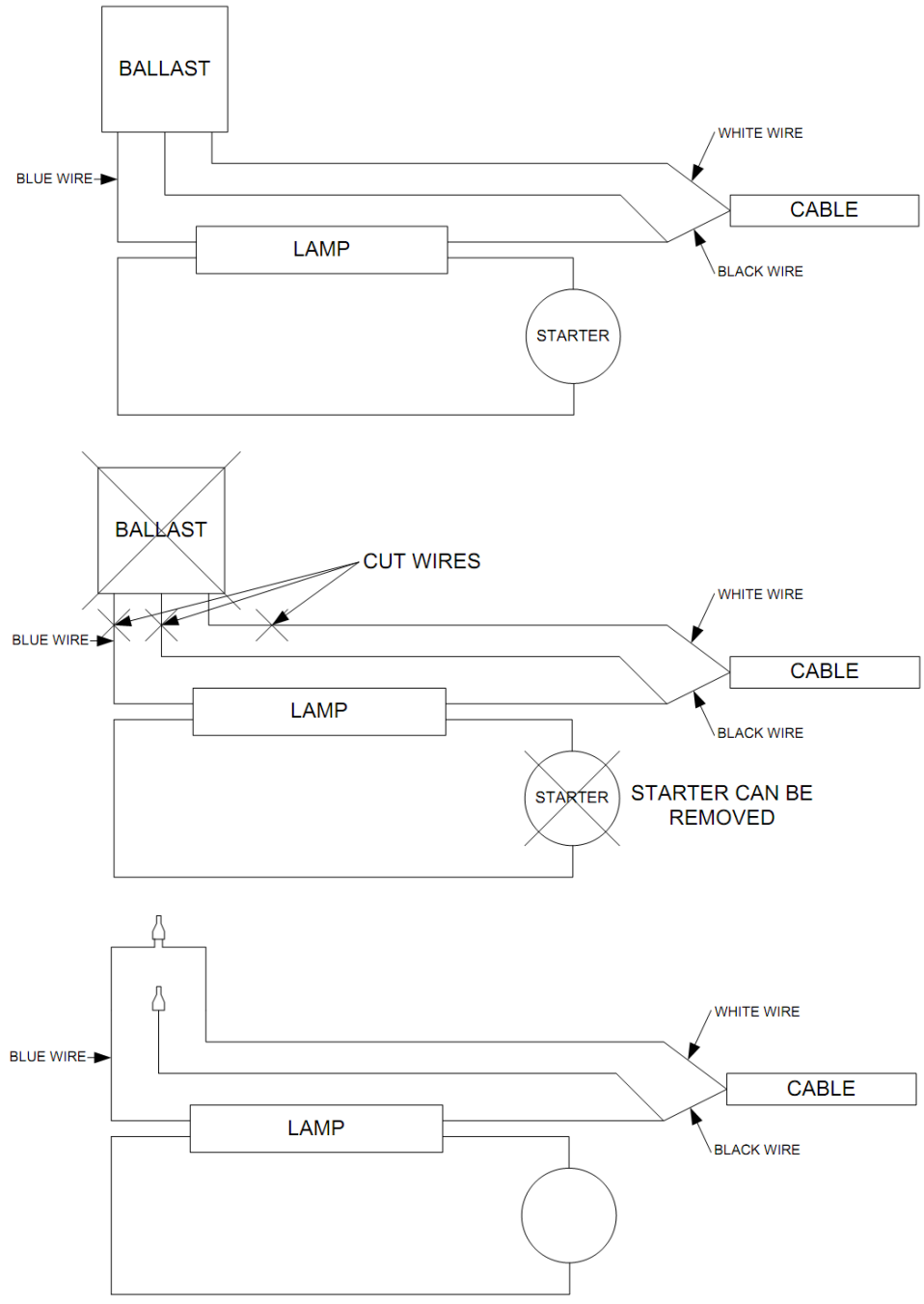


FIGURE 3. Ballast and starter removal for fixtures utilizing M16377/77, 3-wire ballast for 115 VAC operation of solid state lamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.

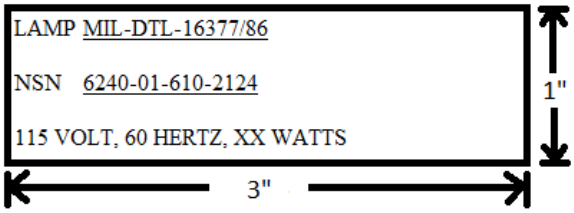


FIGURE 4. Label for lamp.

REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

DESIGN:

Electrical rating:	12 watts, maximum.
Dimensions:	See figure 1.
Nominal bulb diameter:	1½ inches.
Nominal overall length:	24 inches.
Operation:	The solid state lamp shall utilize light emitting diodes (LEDs) and is intended as a replacement of the F20T12 and F17T12 fluorescent lamps used in the following MIL-DTL-16377 fixtures: M16377/8 M16377/11 M16377/12 M16377/57 M16377/73 M16377/74 M16377/75 Colored lighting shall accomplished utilizing existing M16377/42 filters.
Lamp designation:	T12 LED.
Base configuration:	Compatible with medium bipin base of figure SB12 of UL 496 (ANSI G13/G13M).
Weight:	0.95 pounds maximum.
Color rendering index (CRI):	≥75.
Color temperature:	4100 K. See 3.7.4.1.1 of MIL-DTL-16377.
Designed life:	≥50,000 hours.
Average initial lumens:	1050-1400.
Minimum lumens at 50,000 hours (estimated):	750.
Minimum lumens at 0 hours on battery back-up:	240.
Minimum lumen output at 5 hours on battery back-up:	144.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.

Electrical operation:	Lamp shall be compatible with M16377 fixtures utilizing the following ballasts and with the starter removed: M16377/44 M16377/72 Lamp shall also function in M16377 fixtures when the starter has been removed and the ballast has been removed in accordance with figure 2 or figure 3 and ungrounded 115 VAC is applied to the lamp holders. Lamp shall function in New Emergency Automatic Lighting Systems (NEALS) fixtures on battery power.
Circuit design:	The lamp assembly shall incorporate integrated driver circuitry so that no external components are required.
Designation and marking:	One label as shown on figure 4.

EXAMINATION AND TESTS:

First article inspection:	See table I.
Conformance:	
Conformance inspection:	See MIL-DTL-16377.
Comparison inspection:	Same as the first article inspection.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection for complete T12 LED tube. 1/

Tests	Requirement	Test Method	Remarks
Examination	3.13	4.5	2/
Operation	3.13	4.8.1	2/
Dielectric withstanding voltage	3.6.15	4.8.2	2/, 3/
Insulation resistance	3.6.16	4.8.3	2/, 3/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/, 4/
Shock	3.5.12	4.8.7	2/, 3/
Vibration	3.5.13	4.8.8	2/, 3/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/, 4/
Dielectric withstanding voltage	3.6.15	4.8.2	2/, 3/
Insulation resistance	3.6.16	4.8.3	2/, 3/
Noise	3.5.14	4.8.9	2/, 3/
Continuity of grounding	3.6.13.4	4.8.12	2/
Magnetic permeability	3.4.8	4.8.18	2/
Photometrics	3.7.1	4.8.6.1	2/
Brightness	3.7.5	4.8.6.4	2/
Chromaticity	3.7.4	4.8.6.5	2/
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	2/
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit	3.17	4.8.24	
Plastics flammability and smoke	3.4.2.1	--	
Optical uniformity	3.7.7.1	4.8.4	
Electromagnetic interference	3.6.14	4.8.16	2/
Power interface	3.6.18	4.8.21	2/
Power quality	3.6.18.1	4.8.22	2/
Voltage transient spike	3.6.18.2	4.8.23	2/
Emergency light output	3.7.1.1	4.8.6.8	5/
Battery discharge test	3.6.19.3	4.8.26	5/

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ Testing shall be completed in above order.

3/ Testing shall be accomplished with three T12 LED lamps installed in M16377/12 Symbol 333.1 fixture, NSN 6210-00-548-0222.

4/ Watertight.

5/ Testing shall be accomplished with two T12 LED lamps installed in M16377/74 Symbol 77.4EM fixture, NSN 6210-01-477-4615.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.

TABLE II. Standard parts.

Military Part Number M16377/86-	Item Name	National Stock Number (NSN)	Remarks
001	T12 LED Lamp	6240-01-610-2124	1/
NOTE: 1/ 4100 K color temperature.			

Preparing activity:
Navy – SH
(Project 6210-2014-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:56Z
Check the source to verify that this is the current version before use.